

Analysis I

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1 Logic, Sets and Numbers

1.1 Logic

Definition 1.1. A *mathematical Statement* is a well-formulated assertion that is strictly either true or false.

Example 1.2. "Every prime number is odd." is a false statement.

Definition 1.3. A *Statement* that depends on one or multiple variables is called a *predicate* and denoted by $P(x), Q(x), \dots$

Example 1.4.

$$P(x) := x > 2$$

Definition 1.5. The negation of a mathematical Statement A is defined as $\neg A := A$ is false.

Example 1.6. $A := \forall x \in \mathbb{R} : x > 2$ $\neg A := \exists x \in \mathbb{R} : x \leq 2$

Definition 1.7. *Quantifiers* are short hand notation for making statements about predicates.

$$\begin{aligned}\forall x \in \mathbb{R} : P(x) &\iff \text{for all } x \in \mathbb{R} \text{ holds } P(x) \\ \exists x \in \mathbb{R} : P(x) &\iff \text{there exists } x \in \mathbb{R} \text{ s.t. } P(x) \\ \exists! x \in \mathbb{R} : P(x) &\iff \text{there exists exactly one } x \in \mathbb{R} \text{ s.t. } P(x) \\ \nexists x \in \mathbb{R} : P(x) &\iff \text{there exists no } x \in \mathbb{R} \text{ s.t. } P(x)\end{aligned}$$

Example 1.8.

$$\begin{aligned}\forall x \in \mathbb{R} : x^2 &\geq 0 \\ \forall x \in \mathbb{R} \exists y \in \mathbb{R} : x &= y^2\end{aligned}$$

Important 1.9. The order of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.

Definition 1.10. Basic *logical connectives* are defined as:

$$\begin{aligned}A \wedge B &:= A \text{ and } B \text{ are true} \\ A \vee B &:= A \text{ or } B \text{ is true} \\ A \implies B &:= A \text{ implies } B \text{ is true}\end{aligned}$$

$A \iff B := A \text{ is equivalent to } B \text{ is true}$

Remark 1.11. Observe that equivalence holds iff. $A \implies B \wedge B \implies A$

Lemma 1.12. From the above definition it follows that:

$$\neg A \vee B \iff A \implies B$$

This can be shown by evaluating the truth table for both statements.

Theorem 1.13. The following equivalences hold:

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

aswell as

$$\forall x (P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$$

$$\exists x (P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$$

1.2 Sets

Definition 1.14. A **set** is a collection of elements, that are listed or characterized by a certain property. The empty set is denoted by $\emptyset$ and is the only set that contains no elements.

Definition 1.15. An element x is **contained in a set** A iff. $x \in A$ else it is not contained in A denoted by $x \notin A$.

Definition 1.16. Let A and B be sets. A is a **subset** of B denoted by $A \subset B$ iff.

$$\forall x (x \in A \implies x \in B)$$

A is a **proper subset** of B denoted by $A \subsetneq B$ iff.

$$A \subset B \wedge A \neq B$$

Definition 1.17. Let A and B be sets.

- The **union** of A and B is defined as $A \cup B := \{x \mid x \in A \vee x \in B\}$.
- The **intersection** of A and B is defined as $A \cap B := \{x \mid x \in A \wedge x \in B\}$.
- The **difference** of A and B is defined as $A \setminus B := \{x \mid x \in A \wedge x \notin B\}$.
- The **Cartesian product** of A and B is defined as $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.

Remark 1.18. Note that if $B \subset A$ then the difference $A \setminus B$ is called the complement of B in A and is denoted by B^c .

1.3 Numbers

The basic and often used sets of numbers are:

- $\mathbb{N}$: The set of natural numbers 1, 2, 3, 4, ...
- $\mathbb{Z}$: The set of integers ... -2, -1, 0, 1, 2, ...
- $\mathbb{Q}$: The set of rational numbers $\frac{p}{q}$ where $p, q \in \mathbb{Z}$
- $\mathbb{R}$: The set of real numbers

Remark 1.19. Some important subsets of the real numbers are $\mathbb{R}_0^+/\mathbb{R}^+$ and $\mathbb{R}_0^-/\mathbb{R}^-$ which define the intervalls $[0, \infty)/(0, \infty)$ and $(-\infty, 0]/(-\infty, 0)$.

Definition 1.20. Continuous subsets are called **intervalls**. An intervall is called

- **open** if it does not contain its endpoints $I = (a, b) := \{x \mid a < x < b\}$
- **closed** if it contains its endpoints $I = [a, b] := \{x \mid a \leq x \leq b\}$
- **half-open** if it contains one endpoint and not the other $I = [a, b) := \{x \mid a \leq x < b\}$

Remark 1.21. An arbitrary subset of $\mathbb{R}$ is called open if it is a union of open intervalls and closed if its complement is open.

Definition 1.22. An **upper (lower) bound** of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:

$$\forall x \in X : x \leq (\geq) y$$

The **maximum** (**minimum**) of a subset $X \subset \mathbb{R}$ is the element $x_0 \in X$ such that:

$$\forall x \in X : x \leq (\geq) x_0$$

Often denoted as **max**(X) and **min**(X).

Definition 1.23. An interval $I = (a, b)$ is **bounded** if there exist upper and lower bounds.

Remark 1.24. A bounded and closed interval is often referred to as **compact**.

Definition 1.25. The **Supremum** $\sup(X)$ of a subset $X \subset \mathbb{R}$ is the least upper bound of X . and can be characterized by:

$$\forall x \in X : x \leq \sup(X) \wedge \forall \epsilon > 0 : \exists x \in X : x > \sup(X) - \epsilon$$

Definition 1.26. The **Infimum** $\inf(X)$ of a subset $X \subset \mathbb{R}$ is the greatest lower bound of X . and can be characterized by:

$$\forall x \in X : x \geq \inf(X) \wedge \forall \epsilon > 0 : \exists x \in X : x < \inf(X) + \epsilon$$

Example 1.27. Let $X = [-2, 1)$ we have a minimum of $x_0 = -1$ but no maximum. Observe that it is bounded from above by 1 and 2 or any number larger than 1 but only 1 is the least upper bound (Supremum).

Theorem 1.28. The **maximum and minimum are unique** as long as they exist.

Proof. Assume there exists $x_1, x_2 \in X$ such that $x_1 = \max(X)$ and $x_2 = \max(X)$. As both x_1 and x_2 are maximum of X we have

$$x_1 \geq x_2 \wedge x_2 \geq x_1 \implies x_1 = x_2$$

□

Definition 1.29. Axioms of Addition

- *Commutativity:* $x + y = y + x$
- *Associativity:* $x + (y + z) = (x + y) + z$
- *Identity element:* There exists an element $e \in X$ such that $x + e = x$ for all $x \in X$.
- *Inverse element:* For every $x \in X$ there exists an element $x^{-1} \in X$ such that

$$x + x^{-1} = e.$$

Remark 1.30. A set X together with an operation $+$ that satisfies the axioms of addition is called a **abelian group**.

Example 1.31. $(\mathbb{Z}, +)$ is an abelian group.

Definition 1.32. Axioms of Multiplication

- *Commutativity:* $x \cdot y = y \cdot x$
- *Associativity:* $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- *Identity element:* There exists an element $e \in X$ such that $x \cdot e = x$ for all $x \in X$.
- *Inverse element:* For every $x \in X$ there exists an element $x^{-1} \in X$ such that $x \cdot x^{-1} = e$.

Definition 1.33. Distributivity of multiplication over addition:

$$\forall x, y, z \in X : x \cdot (y + z) = x \cdot y + x \cdot z$$

Remark 1.34. A set X together with operations $+$ and $\cdot$ that satisfies the axioms of addition and multiplication and distributivity is called a **field**.

Definition 1.35. Order Axioms

- *reflexivity:* $\forall x \in X : x \leq x$
- *antisymmetry:* $\forall x, y \in X : x \leq y \wedge y \leq x \implies x = y$
- *transitivity:* $\forall x, y, z \in X : x \leq y \wedge y \leq z \implies x \leq z$
- *totality:* $\forall x, y \in X : x \leq y \vee y \leq x$

Definition 1.36. A set X is said to be **(order)complete** iff for every non empty subsets $A, B \subseteq X$

$$\begin{aligned} & \forall a \in A \forall b \in B : a \leq b \\ \implies & \exists c \in X \quad \text{s.t.} \quad \forall a \in A : a \leq c \wedge \forall b \in B : c \leq b \end{aligned}$$

Example 1.37. The real numbers $\mathbb{R}$ satisfy axiomatic properties for addition, multiplication, distributivity and order. Furthermore $\mathbb{R}$ is an order complete field.

Example 1.38. $\mathbb{Q}$ is not order complete. Let $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ and $B = \{x \in \mathbb{Q} \mid x^2 \geq 2\}$. Then $\forall a \in A \forall b \in B : a \leq b$ but there is no $c \in \mathbb{Q}$ such that $c^2 = 2$.

Theorem 1.39. Every non empty, bounded from above (below) subset $X \subset \mathbb{R}$ has a *supremum* (*infimum*).

Proof. Let B be the set of all upper bounds of X . As X is bounded from above and non empty, B is non empty. Therefore

$$\forall x \in X : x \leq b \quad \forall b \in B$$

By the order completeness of $\mathbb{R}$ there exists a $c \in \mathbb{R}$ such that

$$\forall x \in X : x \leq c \wedge \forall b \in B : c \leq b$$

Hence c is the least upper bound of X and therefore $c = \sup(X)$. Assume there exists $c' \in \mathbb{R}$ such that $c' = \sup(X)$ and $c' \neq c$. Then by definition of $\sup(X)$ we have

$$c \leq c' \wedge c' \leq c \implies c = c'$$

□

Theorem 1.40. The *Archimedean Principle* says that

- For $x \in \mathbb{R}$ and $y > 0$ there exists an element $n \in \mathbb{N}$ such that $n * y > x$.
- For every $\epsilon > 0$ there exists an element $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$.

Proof. We prove the first theorem by contradiction. Assume it does not hold hence there exists $x \in \mathbb{R}$ and $y > 0$ such that

$$\forall n \in \mathbb{N} : n * y \leq x$$

This implies that x is an upper bound for the set $X = \{n * y \mid n \in \mathbb{N}\}$. As X is non empty and bounded from above, it has a supremum $c = \sup(X)$. As $y > 0$ $c - y$ can not be an upper bound of X hence there exists $n_0 \in \mathbb{N}$ such that

$$c - y < n_0 * y \implies c < (n_0 + 1) * y$$

This contradicts the assumption that c is the supremum of X .

□

Proof. The second theorem is a direct consequence of the first theorem. It can be shown by setting $x = 1$ in the first theorem. □

Theorem 1.41. The *Young Inequality* states that for all $\epsilon > 0$ and all $x, y \in \mathbb{R}$:

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

Proof.

$$\left(\sqrt{\epsilon}x - \frac{1}{\sqrt{\epsilon}}y \right)^2 \geq 0 \implies \epsilon x^2 - 2xy + \frac{1}{\epsilon} y^2 \geq 0 \implies \epsilon x^2 + \frac{1}{\epsilon} y^2 \geq 2xy$$

□

1.3.1 Vectors and Vector Spaces

Definition 1.42. The *euclidian space* $\mathbb{R}^n$ is the set of all vectors $x = (x_1, x_2, \dots, x_n)$ where $x_i \in \mathbb{R}$.

Definition 1.43. *Addition and scalar multiplication* are defined as:

$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

$$\lambda x = (\lambda x_1, \lambda x_2, \dots, \lambda x_n)$$

Definition 1.44. A *vector space* has to satisfy the following axioms:

- *Commutativity:* $x + y = y + x$
- *Associativity:* $x + (y + z) = (x + y) + z$
- *Identity element:* There exists an element $e \in X$ such that $x + e = x$ for all $x \in X$.
- *Inverse element:* For every $x \in X$ there exists an element $x^{-1} \in X$ such that $x + x^{-1} = e$.
- *Distributivity:* $x \cdot (y + z) = x \cdot y + x \cdot z$

Definition 1.45. For $x, y \in \mathbb{R}^n$, the *standard scalar product* is defined as:

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^n x_i y_i$$

The *euclidean norm* is defined as:

$$\|x\| := \sqrt{\sum_{i=1}^n x_i^2}$$

The *euclidean distance* between x and y is:

$$d(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Theorem 1.46. The triangle- and the cauch-schwarz inequality hold for all $x, y, z \in$

$\mathbb{R}^n$:

$$\begin{aligned} \|x - z\| &\leq \|x - y\| + \|y - z\| \\ |\langle x, y \rangle| &\leq \|x\| \|y\| \end{aligned}$$

1.3.2 Complex Numbers

Definition 1.47. The *imaginary unit* i is defined as $i^2 = -1$.

Definition 1.48. The *complex numbers* $\mathbb{C}$ are the set of numbers

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

where a is the *real part* $Re(z)$, and b is the *imaginary part* $Im(z)$.

Definition 1.49. Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:

- *Addition:* $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$
- *Multiplication:* $z_1 \cdot z_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$

Definition 1.50. The *complex conjugate* of $z = x + iy$ is $\bar{z} = x - iy$.

Important 1.51. Division is calculated by expanding with the conjugate of the denominator:

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

Remark 1.52. Some useful properties of the conjugate:

- $z \cdot \bar{z} = |z|^2$
- $z + \bar{z} = 2Re(z)$ and $z - \bar{z} = 2iIm(z)$
- $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

Definition 1.53. Geometrically, complex numbers can be represented in the Gaussian plane. The *absolute value* of $z = a + ib$ is its *distance from the origin*:

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$.

Remark 1.54. Observe that the triangle inequality holds in $\mathbb{C}$: $|z + w| \leq |z| + |w|$.

Theorem 1.55. The Euler's formula states that for $t \in \mathbb{R}$:

$$e^{it} = \cos(t) + i \sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Remark 1.56. The *Intuition* behind eulers formula is we can use Taylor series to show that:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Using $x = it$ we get:

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \cdots = \cos(t) + i \sin(t)$$

Eulers formula leads us to a new representation of complex numbers: **the polarcoordinates**.

Definition 1.57. For $z \in \mathbb{C}$, the polar-coordinates are defined as:

$$z = r(\cos(t) + i \sin(t)) = re^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Remark 1.58. To convert from cartesian to polar coordinates:

$$r = \sqrt{a^2 + b^2}, \quad t = \arctan\left(\frac{b}{a}\right)$$

Note that the arctan function only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so we have to adjust the angle based on the quadrant of the complex number.

Example 1.59. Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1+t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1 - t_2)}$$

Example 1.60. *So is exponentiation or taking roots:*

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i \frac{t+2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

Definition 1.61. *The n -th roots of unity are the solutions to $z^n = 1$, which are given by:*

$$z_k = e^{i \frac{2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

Theorem 1.62. *The sum of the roots of unit is always zero.*

$$\sum_{k=0}^{n-1} e^{i \frac{2\pi k}{n}} = 0$$

Theorem 1.63. *The **fundamental theorem of algebra** states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.*

2 Sequences and Series

2.1 Sequences

Definition 2.1. A **sequence** (Folge) $(a_n)_{n \in \mathbb{N}_0}$ is an infinite list of numbers, mathematically a mapping $f : \mathbb{N}_0 \rightarrow X$. It can be given explicitly or recursively as

$$a_n = f(n) \quad \text{or} \quad a_n = g(a_{n-1}, a_{n-2}, \dots)$$

Example 2.2. A famous example is the Fibonacci sequence defined by $a_0 = 0, a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. The first few terms are 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

Definition 2.3. A sequence (a_n) is called **convergent** with limit $L \in \mathbb{R}$ if:

$$\forall \epsilon > 0 \exists N > 0 \quad \text{s.t.} \quad \forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n \rightarrow \infty} a_n = L$. If no such L exists, the sequence is **divergent**.

Remark 2.4. A special case of divergent sequences are sequences which diverge to $\pm\infty$. Their limits are called **improper limits**.

Theorem 2.5. The limit of a sequence is unique as long as it exists.

Proof. Assume (a_n) converges to L_1 and L_2 . Then for any $\epsilon > 0$ there exists N_1 and N_2 such that:

$$|a_n - L_1| < \epsilon \quad \forall n > N_1$$

$$|a_n - L_2| < \epsilon \quad \forall n > N_2$$

Then for any $n > \max(N_1, N_2)$:

$$|L_1 - L_2| = |L_1 - a_n + a_n - L_2| \leq |L_1 - a_n| + |a_n - L_2| < 2\epsilon$$

Since this holds for any $\epsilon > 0$, we must have $L_1 = L_2$. □

Theorem 2.6. Any subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ of a convergent sequence converges to the same limit L .

Definition 2.7. An **accumulation point** (Häufungspunkt) $A \in \mathbb{R}$ of a sequence fulfills:

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \epsilon$$

Example 2.8. The sequence $a_n = \frac{1}{2}(-1)^n$ has the accumulation points $A_1 = \frac{1}{2}$ and $A_2 = -\frac{1}{2}$.

Theorem 2.9. *A is an accumulation point iff there exists a convergent subsequence with limit A .*

Proof. Let (a_n) be a sequence and A an accumulation point. We can easily construct a subsequence (a_{n_k}) that converges to A by selecting n_k such that $|a_{n_k} - A| < 1/k$ this is given to exist by the definition of an accumulation point. Now let (a_{n_k}) be a convergent subsequence with limit A then for any $\epsilon > 0$ there exists K such that $|a_{n_k} - A| < \epsilon$ for all $k > K$ and since $n_k \rightarrow \infty$ as $k \rightarrow \infty$ by definition it follows that A is an accumulation point. $\square$

Theorem 2.10. *Every convergent sequence has exactly one accumulation point (its limit).*

Proof. For A to be an accumulation point of a_n we need that there are indefinitely many a_n that approach A . If a_n converges to L then we can intuitively say that that A has to be L . $\square$

Definition 2.11. *A sequence is **upper (lower) bounded** if there exists M such that $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.*

Theorem 2.12. *Every convergent sequence is bounded.*

Proof. By the definition of convergence there exists N such that for all $n \geq N$ we have $|a_n - L| < 1$. Thus $|a_n| < |L| + 1$ for all $n \geq N$. Further we have a finite number of terms $a_0, \dots, a_{N-1}$ which are also bounded. Therefore the sequence is bounded by $M = \max(|a_0|, \dots, |a_{N-1}|, |L| + 1)$. $\square$

Theorem 2.13. Bolzano-Weierstrass Theorem: *Every bounded sequence has an accumulation point.*

Theorem 2.14. Limit arithmetic: *For $\lim a_n = K$, $\lim b_n = L$ and $C \in \mathbb{R}$:*

- $\lim(a_n \pm b_n) = K \pm L$
- $\lim(C \cdot a_n \cdot b_n) = C \cdot K \cdot L$
- $\lim(a_n/b_n) = K/L$ (if $L \neq 0$ and $b_n \neq 0$)

Remark 2.15. Important limits:

- $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

- $\lim_{n \rightarrow \infty} n^{1/n} = 1$
- $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$

Theorem 2.16. Sandwich-Theorem: If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.

Proof. Let $\lim a_n = \lim c_n = L$ and $\lim b_n = M$. From the above arithmetic we can see that $a_n \leq b_n \leq c_n$ implies $L \leq M \leq L$ so we have that b_n is squeezed between a_n and c_n and thus must converge to $M = L$. $\square$

Example 2.17. We want to calculate $\lim_{n \rightarrow \infty} (2^n + 5^n)^{\frac{1}{n}}$. We can bound the sequence by $5^n \leq 2^n + 5^n \leq 2 \cdot 5^n$. Taking the n -th root we get $5 \leq (2^n + 5^n)^{\frac{1}{n}} \leq 2^{\frac{1}{n}} \cdot 5$. As $n \rightarrow \infty$ we have $2^{\frac{1}{n}} \rightarrow 1$ so by the sandwich theorem we have $\lim_{n \rightarrow \infty} (2^n + 5^n)^{\frac{1}{n}} = 5$.

Remark 2.18. The sandwich theorem is often used to find upper and lower bounds for a sequence that converge to the same limit and therefore conclude that the original sequence does too.

Definition 2.19. A sequence is **monotonically increasing (decreasing)** if $a_n \leq (\geq) a_m$ for $m > n$.

Theorem 2.20. Monotone Convergence Theorem: Every bounded monotonic sequence converges to its supremum (infimum).

Proof. Wlog let (a_n) be a bounded monotonically increasing sequence. Hence there exists $S = \sup\{a_n : n \in \mathbb{N}_0\}$ such that $a_n \leq S$ for all n and for any $\epsilon > 0$ there exists N such that $a_N > S - \epsilon$ and therefore $|a_n - S| < \epsilon$ for all $n \geq N$. $\square$

Definition 2.21. The limit of the supremum is called **limes superior** and denoted as:

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$$

The limit of the infimum is called **limes inferior** and denoted as:

$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k$$

Definition 2.22. A sequence is a **Cauchy sequence** iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \epsilon$$

Theorem 2.23. Every Cauchy sequence is bounded.

Proof. Choose $\epsilon = 1$, then there exists N such that $|a_n - a_N| < 1$ for all $n \geq N$. Thus $|a_n| < |a_N| + 1$ for $n \geq N$, meaning (a_n) is bounded by $M = \max(|a_0|, \dots, |a_{N-1}|, |a_N| + 1)$. $\square$

Theorem 2.24. A sequence of real numbers converges iff it is a Cauchy sequence.

Proof. $\implies$: Let (a_n) converge to L . For any $\epsilon > 0$, there exists N such that $\forall n \geq N : |a_n - L| < \epsilon/2$. Then for any $m, n \geq N$:

$$|a_n - a_m| = |a_n - L + L - a_m| \leq |a_n - L| + |L - a_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, (a_n) is a Cauchy sequence.

$\impliedby$: Let (a_n) be a Cauchy sequence therefore it is bounded. By the Bolzano-Weierstrass theorem, the bounded sequence (a_n) has an accumulation point L , so there exists a subsequence (a_{n_k}) converging to L . For any $\epsilon > 0$, since (a_n) is Cauchy, there exists N' such that $\forall m, n \geq N' : |a_n - a_m| < \epsilon/2$. Since $a_{n_k} \rightarrow L$, choose a k such that $n_k \geq N'$ and $|a_{n_k} - L| < \epsilon/2$. Then for any $n \geq N'$:

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, the sequence converges to L . $\square$

2.2 Series

Definition 2.25. A **series** (Reihe) is an infinite sum $\sum_{i=0}^{\infty} a_i$ of the terms of a sequence (a_n) .

Definition 2.26. A series is said to converge iff the sequence of partial sums converges.

$$\sum_{i=0}^{\infty} a_i = \lim_{n \rightarrow \infty} \sum_{i=0}^n a_i$$

Remark 2.27. Some examples of series are:

- **geometric series:** $\sum_{n=0}^{\infty} x^n$ converges to $\frac{1}{1-x}$ iff $|x| < 1$
- **harmonic series:** $\sum_{n=1}^{\infty} \frac{1}{n}$ converges for $s > 1$

Theorem 2.28. Necessary condition for convergence: for $\sum_{i=0}^{\infty} a_i$ to converge we need $\lim_{n \rightarrow \infty} a_n = 0$ i.e. it is a null sequence.

Theorem 2.29. For series with non-negative terms $a_n \geq 0$, the sequence of partial sums is monotonically increasing and converges iff it is bounded.

Definition 2.30. A series is **absolutely convergent** if $\sum |a_n|$ converges. A **conditionally convergent** series converges, but not absolutely.

Remark 2.31. Absolute convergence implies normal convergence.

Theorem 2.32. Riemann's rearrangement theorem: Let $\sum a_n$ be a conditionally convergent series. Then for any $x \in \mathbb{R}$, there exists a rearrangement of the series that converges to x .

Theorem 2.33. Comparison test (Vergleichskriterium): If $c_n \leq b_n \leq a_n$ for $n \geq N$:

- If $\sum a_n$ converges, then $\sum b_n$ converges (**Majorantenkriterium**).
- If $\sum c_n$ diverges, then $\sum b_n$ diverges (**Minorantenkriterium**).

Example 2.34.

$$\sum_{n=1}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots$$

We can compare this series to the geometric series

$$1 + \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = 1 + 2 = 3$$

Therefore according to the Majorantenkriterium the series $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

Theorem 2.35. Leibniz Criterion: For an alternating series $\sum (-1)^n a_n$ with a monotonically decreasing null sequence (a_n) , the series converges.

Proof. Let (a_n) be a non-negative, monotonically decreasing sequence converging to 0. Let $s_n = \sum_{k=0}^n (-1)^k a_k$ be the n -th partial sum. For the even partial sums (s_{2n}) , we have

$$s_{2n+2} - s_{2n} = -a_{2n+1} + a_{2n+2} \leq 0$$

since $a_{2n+2} \leq a_{2n+1}$. Thus, (s_{2n}) is monotonically decreasing. Similarly for the odd partial sums (s_{2n+1}) , we have

$$s_{2n+3} - s_{2n+1} = a_{2n+2} - a_{2n+3} \geq 0$$

meaning (s_{2n+1}) is monotonically increasing. Note that

$$s_{2n+1} = s_{2n} - a_{2n+1} \leq s_{2n} \leq s_0 = a_0$$

This limits both bounds: $0 \leq s_{2n+1} \leq s_{2n} \leq a_0$. By the Monotone Convergence Theorem, both bounded monotonic subsequences converge: $\lim_{n \rightarrow \infty} s_{2n} = L_1$ and $\lim_{n \rightarrow \infty} s_{2n+1} = L_2$. Finally,

$$L_1 - L_2 = \lim_{n \rightarrow \infty} (s_{2n} - s_{2n+1}) = \lim_{n \rightarrow \infty} a_{2n+1} = 0$$

So $L_1 = L_2 = L$. Since both the even and odd subsequences converge to the same limit L , the entire sequence of partial sums (s_n) converges to L , thus the series converges. $\square$

Theorem 2.36. Cauchy Criterion: The series $\sum a_n$ converges iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : \left| \sum_{k=n}^m a_k \right| < \epsilon$$

Theorem 2.37. Root Criterion (*Wurzelkriterium*): Let $\rho = \limsup \sqrt[n]{|a_n|}$. If $\rho < 1$, abs conv; if $\rho > 1$, div.

Theorem 2.38. Ratio Criterion (*Quotientenkriterium*): Let $\rho = \lim \left| \frac{a_{n+1}}{a_n} \right|$. If $\rho < 1$, abs conv; if $\rho > 1$, div.

Example 2.39.

$$\sum_{n=1}^{\infty} n^4 e^{-n}$$

$$\rho = \lim_{n \rightarrow \infty} \frac{(n+1)^4 e^{-(n+1)}}{n^4 e^{-n}} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^4 e^{-1} = e^{-1} < 1$$

Therefore the series converges.

Definition 2.40. A *power series* (Potenzreihe) centered at a has the form

$$\sum_{k=0}^{\infty} c_k (x - a)^k$$

Theorem 2.41. Every power series has a **convergence radius** R , with the **interval of convergence** $(a - R, a + R)$. R is computed by the formula $R = \frac{1}{\limsup |c_n|^{1/n}}$.

Example 2.42.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} x^n$$

$$\rho = \limsup \frac{1}{n^{1/n}} = 1 \implies R = 1$$

Therefore the series converges for

$$x \in (-1, 1)$$

.

Remark 2.43. For $x = 1$ we have the harmonic series, which diverges. For $x = -1$ we have the alternating harmonic series, which converges.

3 Functions

Definition 3.1. A **function** $f : D \rightarrow C$ is a map from a domain D to a codomain C s.t. for each $x \in D$ there is exactly one $y \in C$ st. $f(x) = y$.

Remark 3.2. The input is called the **independent variable** (argument) while the output is called the **dependent variable**.

Definition 3.3. A function f is called **bounded** if there exists $M \in \mathbb{R}$ such that $|f(x)| \leq (\geq) M$ for all $x \in D$.

Definition 3.4. A function f is called

- **injective** iff $\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$.
- **surjective** iff $\forall y \in B \exists x \in A : f(x) = y$.
- **bijective** iff f is injective and surjective.

Example 3.5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^2$. Observe that f is not surjective as no output is negative, nor is it injective as for ex. $f(1) = f(-1) = 1$. But $g : \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$ with $g(x) = x^2$ is both injective and surjective.

Definition 3.6. The **restriction** of f under $D' \subset \mathbb{D}$ is defined as $f|_{D'} : D' \rightarrow C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$.

Remark 3.7. Note that $f|_{D'}$ and f are two different functions.

Definition 3.8. Let $f : D \rightarrow C$ and $g : C \rightarrow B$ be functions then

$$g \circ f : D \rightarrow B \quad \text{with} \quad g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the **composition** of f and g .

Definition 3.9. The **identity function** $id_D : D \rightarrow D$ is defined as $id_D(x) = x \quad \forall x \in D$.

Definition 3.10. If a function $f : D \rightarrow C$ is bijective then there exists a unique function $f^{-1} : C \rightarrow D$ such that

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C$$

and

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

f^{-1} is called the **inverse function** of f .

Definition 3.11. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called

- **even** iff $\forall x \in \mathbb{R} : f(-x) = f(x)$
- **odd** iff $\forall x \in \mathbb{R} : f(-x) = -f(x)$

Example 3.12. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^3$ is odd as $f(-x) = (-x)^3 = -x^3 = -f(x)$ while the function $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(x) = x^2$ is even as $g(-x) = (-x)^2 = x^2 = g(x)$.

Definition 3.13. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called

- **(strictly) monotonically increasing** iff $\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < f(x_2)$
- **(strictly) monotonically decreasing** iff $\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) > f(x_2)$
- **(strictly) monotonic** iff f is either (strictly) monotonically increasing or (strictly) monotonically decreasing

Theorem 3.14. Every strictly monotonic function is injective.

Proof. Let f be strictly monotonic (wlog increasing) and assume it is not injective then there exists $x_1 \neq x_2$ such that $f(x_1) = f(x_2)$. Wlog assume $x_1 < x_2$ then $f(x_1) < f(x_2)$ which contradicts the assumption that $f(x_1) = f(x_2)$. $\square$

Remark 3.15. Observe that functions do not have to be even or odd.

Definition 3.16. The **graph** of a function $f : D \rightarrow C$ is the set of all ordered pairs $(x, f(x))$ for $x \in D$. It is **convex** if it lies below the line segment connecting any two points on the graph. Similarly it is called **concave** if it lies above the line segment connecting any two points on the graph.

3.1 Limits in the context of functions

Definition 3.17. Let $f : \mathbb{D} \rightarrow \mathbb{R}$ be a function satisfying $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for all $\delta > 0$ and $x_0 \in \mathbb{R}$. We say that $\lim_{x \rightarrow x_0} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t.}$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$

The limit L is unique.

Remark 3.18. The intuition behind the definition of the limit is that for any given distance $\varepsilon > 0$ around L , we can find a $\delta > 0$ such that for all $x \in \mathbb{D}$ that are at least δ close to x_0 , we have $|f(x) - L| < \varepsilon$. So for any bound on the output (ε) there exists a bound on the input (δ) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(L - \varepsilon, L + \varepsilon)$.

Definition 3.19. A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has in x_0 a

- **right-sided limit** $\lim_{x \searrow x_0} f(x) = \lim_{x \rightarrow x_0^+} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } \forall x \in \mathbb{D} (x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

- **left-sided limit** $\lim_{x \nearrow x_0} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } \forall x \in \mathbb{D} (x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon)$$

Definition 3.20. A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has a

- **limit for x approaching infinity** $\lim_{x \rightarrow \infty} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists M > 0 \text{ s.t. } \forall x \in \mathbb{D} (x > M \implies |f(x) - L| < \varepsilon)$$

- **limit for x approaching negative infinity** $\lim_{x \rightarrow -\infty} f(x) = L$ iff

$$\forall \varepsilon > 0 \exists M > 0 \text{ s.t. } \forall x \in \mathbb{D} (x < -M \implies |f(x) - L| < \varepsilon)$$

Definition 3.21. A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has in x_0 a

- **improper limit approaching infinity** $\lim_{x \rightarrow x_0} f(x) = \infty$ iff

$$\forall N > 0 \exists \delta > 0 \text{ s.t. } \forall x \in \mathbb{D} (0 < |x - x_0| < \delta \implies f(x) > N)$$

- **improper limit approaching negative infinity** $\lim_{x \rightarrow x_0} f(x) = -\infty$ iff

$$\forall N > 0 \exists \delta > 0 \quad \text{s.t.} \quad \forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) < -N)$$

Theorem 3.22. Let $A \in \mathbb{R}$ and assume $\lim_{x \rightarrow x_0} f(x) = K (\neq \pm\infty)$ and $\lim_{x \rightarrow x_0} g(x) = L (\neq \pm\infty)$.

- $\lim_{x \rightarrow x_0} (f(x) + g(x)) = K + L$
- $\lim_{x \rightarrow x_0} (f(x) - g(x)) = K - L$
- $\lim_{x \rightarrow x_0} (f(x)g(x)) = KL$
- $\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L$
- $\lim_{x \rightarrow x_0} (Af(x)) = AK$
- $\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L$ if $L \neq 0$

3.2 Continuity of functions

Definition 3.23. A function $f : D \rightarrow \mathbb{R}$ is called **continuous** at x_0 if

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that } \forall x \in I(|x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon)$$

The function is called **continuous** if it is continuous at every point in its domain.

Remark 3.24. The **intuition** behind this definition is that for any given distance $\epsilon > 0$ around $f(x_0)$, we can find a $\delta > 0$ such that for all $x \in D$ that are at least δ close to x_0 , we have $|f(x) - f(x_0)| < \epsilon$. So for any bound on the output (ϵ) there exists a bound on the input (δ) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(f(x_0) - \epsilon, f(x_0) + \epsilon)$.

Definition 3.25. Let $f : D \rightarrow \mathbb{R}$ be a function. Then f is called **left(right)-continuous** at x_0 if

$$\lim_{x \rightarrow x_0^{-(+)}} f(x) = f(x_0)$$

Theorem 3.26. Let $f : D \subset \mathbb{D}(f) \subset \mathbb{R}$ be a function and $x_0 \in D$. f is continuous at x_0 iff for every sequence $(x_n)_{n \in \mathbb{N}} \subset D$ with $\lim_{n \rightarrow \infty} x_n = x_0$ it holds that $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$.

Remark 3.27. Continuity can also be viewed in terms of intervals, where a function f is continuous iff for every open interval $I \subset \mathbb{R}$ it holds that $f^{-1}(I)$ is open.

Lemma 3.28. Let f, g be continuous functions, the following holds:

- $f + g$ is continuous
- $f - g$ is continuous
- $f \cdot g$ is continuous
- $c \cdot f$ is continuous for all $c \in \mathbb{R}$
- $\frac{f}{g}$ is continuous if $g(x) \neq 0$
- $f \circ g$ is continuous

Proof. Addition/Subtraction/Multiplication/Division follow from the sequence-definition of continuity and the properties of limits. It is left to prove that $f \circ g$ is continuous if f and g are. Let g be continuous at a point c , and let f be continuous at the point $g(c)$.

By the limit definition of continuity, since g is continuous at c , we know:

$$\lim_{x \rightarrow c} g(x) = g(c)$$

Similarly, since f is continuous at $g(c)$, for any variable y approaching $g(c)$, we know:

$$\lim_{y \rightarrow g(c)} f(y) = f(g(c))$$

To find the limit of the composite function $f(g(x))$ as $x \rightarrow c$, we substitute $g(x)$ for our input y . Because f is continuous, we are allowed to pass the limit directly inside the function f :

$$\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right)$$

Now, we just substitute the known limit of $g(x)$:

$$f\left(\lim_{x \rightarrow c} g(x)\right) = f(g(c))$$

Therefore, $\lim_{x \rightarrow c} (f \circ g)(x) = (f \circ g)(c)$. This perfectly satisfies the definition of continuity, proving that $f \circ g$ is continuous at c □

Theorem 3.29. Let I be an interval and let $f : I \rightarrow J$ be a continuous and bijective function. Then J is an interval too and the inverse function $f^{-1} : J \rightarrow I$ is continuous.

Theorem 3.30. *Let $f : D \rightarrow \mathbb{R}$ be a continuous function. Let $c \in [f(a), f(b)]$ then there exists a $x_0 \in [a, b]$ such that $f(x_0) = c$.*

Proof. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Without loss of generality, assume $f(a) < f(b)$. We choose an arbitrary $c \in \mathbb{R}$ such that $f(a) \leq c \leq f(b)$ and define the set:

$$X := \{t \in [a, b] \mid f(t) \leq c\}$$

Observe that $X \neq \emptyset$ (since $a \in X$ as $f(a) \leq c$) and X is bounded above by b . Thus, by the completeness axiom of the real numbers, the supremum $s := \sup X$ exists, and clearly $s \in [a, b]$.

By the definition of the supremum, for all $n \in \mathbb{N}$, there exists $t_n \in X \cap [s - \frac{1}{n}, s]$. Because $t_n \rightarrow s$ as $n \rightarrow \infty$ and f is continuous, we have $\lim_{n \rightarrow \infty} f(t_n) = f(s)$. Since $t_n \in X$, we know $f(t_n) \leq c$ for all $n \in \mathbb{N}$. By the limit-preserving properties of inequalities, it follows that $f(s) \leq c$.

Now, assume for the sake of contradiction that $f(s) < c$. Since $f(s) < c \leq f(b)$, we know $s \neq b$, which implies $s < b$.

Let $\epsilon = c - f(s) > 0$. By the continuity of f at s , there exists a $\delta > 0$ such that for all $y \in [a, b]$:

$$|y - s| < \delta \implies |f(y) - f(s)| < \epsilon$$

This implies $f(y) < f(s) + \epsilon = c$.

Now, define $y := \min(b, s + \frac{\delta}{2})$. Since $s < b$ and $\delta > 0$, we strictly have $y > s$. Furthermore, $y \in [a, b]$ and $|y - s| = \frac{\delta}{2} < \delta$. Therefore, by our continuity implication, $f(y) < c$, which means $y \in X$.

However, we have found an element $y \in X$ such that $y > s$, which directly contradicts the fact that s is an upper bound (the supremum) of X .

Thus, our assumption that $f(s) < c$ must be false. Since we have already established that $f(s) \leq c$, we conclude that $f(s) = c$. $\square$

Definition 3.31. *A bounded and closed interval is called **compact**.*

Example 3.32. *The interval $[0, 1]$ is compact while $[0, \infty)$ is not.*

Lemma 3.33. *Let $[a, b]$ be a compact interval and let $(x_n)_{n \in \mathbb{N}} \subset [a, b]$ be a sequence. Then there exists a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ such that*

$$\lim_{k \rightarrow \infty} x_{n_k} = x_0 \text{ for some } x_0 \in [a, b].$$

Definition 3.34. *Let $f : D \subset \mathbb{R} \rightarrow \mathbb{R}$ be a function*

- *If for $x_0 \in D$ it holds that $f(x_0) \geq f(x) \forall x \in D$, then f has a local maximum at x_0 .*

- If for $x_0 \in D$ it holds that $f(x_0) \leq f(x) \forall x \in D$, then f has a local minimum at x_0 .
- Local maxima and minima are called local extrema.

Theorem 3.35. *Let $f : I \subset \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and I be a compact interval. Then f is bounded and attains its maximum and minimum value on I . Hence*

$$\exists x_{\min}, x_{\max} \in I \text{ such that } f(x_{\min}) \leq f(x) \leq f(x_{\max}) \forall x \in I$$

4 Differential calculus

Definition 4.1. The *average rate of change* of a function f in the domain between a and $a + h$ is defined as

$$\frac{\Delta y}{\Delta x} = \frac{f(a + h) - f(a)}{h}$$

and also called the *difference quotient*.

Definition 4.2. The *derivative* f' of a function f at the coordinate a is defined as

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} = \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a} = \frac{df}{dx}|_{x=a} = f'(a)$$

Remark 4.3. The derivative $f'(a)$ can geometrically interpreted as the incline/decline of the graph of f at the point $(a, f(a))$.

Definition 4.4. If for a function f at the coordinate a the derivative $f'(a)$ exists then f is called *differentiable* at a . If f is differentiable for every $a \in \mathbb{D}$ then f is called *differentiable*.

Example 4.5. let $f(x) = |x|$ we can see that for $a = 0$ there is no derivative (This can be seen in the graph as a "sharp" corner)

Remark 4.6. We can generally expect that there are no derivatives at undefined points $a \notin \mathbb{D}$.

Lemma 4.7. If f is differentiable at x_0 then f is continuous at x_0 .

Proof. Assume f differentiable then

$$\begin{aligned} \lim_{x \rightarrow x_0} f(x) &= \lim_{h \rightarrow 0} f(x_0 + h) = \lim_{h \rightarrow 0} f(x_0 + h) - f(x_0) + f(x_0) \\ &= f(x_0) + \lim_{h \rightarrow 0} f(x_0 + h) - f(x_0) = f(x_0) + \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \cdot h \\ &= f(x_0) + \lim_{h \rightarrow 0} h \cdot \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \end{aligned}$$

Observe that the last term exists by the assumption hence

$$= f(x_0) + 0 \cdot f'(x_0) = f(x_0)$$

□

Definition 4.8. If the limit

$$\lim_{h>0 \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

existst then we call it the **right-sided derivative** (or the left-sided derivative analogue)

Definition 4.9. For a function f we can iteratively define higher derivatives for $n \in \mathbb{N}_0$

$$f^{(0)} = f, f^{(1)} = f', f^{(2)} = f'', \dots f^{(n+1)} = (f^{(n)})'$$

if $f^{(n)}$ exists we call it **n -times differentiable**.

Definition 4.10. if $f^{(n)}$ is continous we call f **n -times continous differentiable**. The set of all n -times continous differentiable functions on D is denoted as $C^n(D)$

Definition 4.11. We call $C^\infty(D) := \cap_{n=0}^\infty C^n(D)$ the set of **smooth functions**.

Definition 4.12. If for a function f and some $x_0 \in \mathbb{D}$ and $\delta > 0$ we have

$$f(x_0) \geq (\leq) f(x) \quad \forall x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta)$$

then x_0 is called a **local maximum (minimum)**.

4.1 Derivation rules

Theorem 4.13. For functions $f(x)$ we have

- $f(x) = ax^p \implies f'(x) = apx^{p-1}$
- $f(x) = a^x \implies f'(x) = \ln(a) \cdot a^x$
- $f(x) = \sin(x) \implies f'(x) = \cos(x)$
- $f(x) = \cos(x) \implies f'(x) = -\sin(x)$
- $f(x) = \tan(x) \implies f'(x) = \frac{1}{\cos^2(x)} = 1 + \tan^2(x)$
- $f(x) = \sinh(x) \implies f'(x) = \cosh(x)$
- $f(x) = \cosh(x) \implies f'(x) = \sinh(x)$
- $f(x) = \ln(x) \implies f'(x) = \frac{1}{x}$

- $f(x) = \arcsin(x) \implies f'(x) = \frac{1}{\sqrt{1-x^2}}$
- $f(x) = \arccos(x) \implies f'(x) = \frac{-1}{\sqrt{1-x^2}}$
- $f(x) = \arctan(x) \implies f'(x) = \frac{1}{1+x^2}$

Theorem 4.14. Sum Rule: For the n -th derivative of a sum:

$$(f + g)^{(n)} = f^{(n)} + g^{(n)}$$

Proof. This can be proved inductively by applying the definition of the derivative n times.

$$\begin{aligned} (f + g)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) + g(x+h) - f(x) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= f'(x) + g'(x) \end{aligned}$$

For $n = 2$ we have

$$(f + g)''(x) = (f' + g')'(x) = f''(x) + g''(x)$$

and so on. □

Theorem 4.15. General Leibniz Rule (Product): The n -th derivative of a product is given by:

$$(f \cdot g)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}$$

Proof. This is proved analogously to the sum rule.

$$\begin{aligned} (f \cdot g)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{g(x+h)(f(x+h) - f(x))}{h} + \lim_{h \rightarrow 0} \frac{f(x)(g(x+h) - g(x))}{h} \\ &= g(x)f'(x) + f(x)g'(x) \end{aligned}$$

For $n = 2$ we have

$$(f \cdot g)''(x) = (f'g + fg')'(x) = f''g + f'g' + f'g' + fg'' = f''g + 2f'g' + fg''$$

and so on. □

Theorem 4.16. Chain Rule: For the composition of functions:

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

Proof. Observe that

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + g(y) - g(y_0) - g'(y_0)(y - y_0)$$

with the first two summands being a linear (tangential) approximation of $g(y)$ near y_0

$$g(y) \approx g(y_0) + g'(y_0)(y - y_0)$$

By taking out $(y - y_0)$ of the last two summands in the original equation we get:

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + \left[\frac{g(y) - g(y_0)}{y - y_0} - g'(y_0) \right] (y - y_0)$$

We define $\epsilon = \frac{g(y) - g(y_0)}{y - y_0} - g'(y_0)$ and get

$$\begin{aligned} & \lim_{x \rightarrow x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0} \\ &= \lim_{x \rightarrow x_0} (f(f(x_0)) + g'(f(x_0))[f(x) - f(x_0)] + \epsilon f(x)[f(x) - f(x_0)] - g(f(x_0))) \frac{1}{x - x_0} \\ &= g'(y_0)f'(x_0) \end{aligned}$$

□

Theorem 4.17. Quotient Rule: For $g(x) \neq 0$:

$$\left(\frac{f}{g} \right)' = \frac{f'g - fg'}{g^2}$$

Remark 4.18. This follows directly from the product and the chain rule.

Theorem 4.19. Inverse Functions: If f is differentiable and injective:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

4.2 Bernoulli-L'Hopital

Theorem 4.20. Let $I \subset \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$, and let x_0 be a local Extrema of f . Then one of the following holds:

- x_0 is an endpoint of I
- f is not differentiable at x_0

- f is differentiable at x_0 and $f'(x_0) = 0$

Remark 4.21. From $f'(x_0) = 0$ we can not conclude that x_0 is an extremum.

Theorem 4.22. Rolle's Theorem: Let $a < b$ with $f(a) = f(b)$ and $f : [a, b] \rightarrow \mathbb{R}$ a completely continuous function on $[a, b]$ and differentiable on (a, b) then $\exists \xi \in (a, b)$ with $f'(\xi) = 0$

Proof. As $[a, b]$ is compact we can conclude that there exists ξ such that $f(\xi)$ is max or min hence $f'(\xi) = 0$. If $\xi \in (a, b)$ we found our ξ else the max and min is at a therefore also at b hence the function is constant which implies that $f'(x) = 0 \forall x \in [a, b]$. $\square$

Theorem 4.23. Mean-Value-Theorem: Let $a < b, f : [a, b] \rightarrow \mathbb{R}$ a completely continuous function on $[a, b]$ and differentiable on (a, b) then $\exists \xi \in (a, b)$ with

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

Proof. We define a function g with

$$g(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$$

and observe that g fulfills the prerequisites to apply Rolle's theorem. Hence

$$\begin{aligned} \exists \xi g'(\xi) &= 0 \\ \implies \exists \xi f'(\xi) - \frac{f(b) - f(a)}{b - a} \xi &= 0 \\ \implies \exists \xi f'(\xi) &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

$\square$

Theorem 4.24. Cauchy's Mean-Value-Theorem: Let $a < b, f : [a, b] \rightarrow \mathbb{R}$ a completely continuous function on $[a, b]$ and differentiable on (a, b) then $\exists \xi \in (a, b)$ with

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

Proof. We define a function

$$F(x) := g(x)(f(b) - f(a)) - f(x)(g(b) - g(a))$$

and observe that F fulfills the prerequisites to apply Rolle's theorem. Hence

$$\begin{aligned} & \exists \xi F'(\xi) = 0 \\ \implies & \exists \xi g'(\xi)(f(b) - f(a)) - f'(\xi)(g(b) - g(a)) = 0 \\ \implies & \exists \xi \frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)} \end{aligned}$$

□

Theorem 4.25. Bernoulli-L'Hopital's Rule: Let $f, g : I \rightarrow \mathbb{R}$ be differentiable functions on (a, b) and let $x_0 \in (a, b)$. If $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$. If

$$\lim_{x \rightarrow x_0} |f(x)| = \lim_{x \rightarrow x_0} |g(x)| \in \{0, \infty\} \quad \text{and} \quad \exists L = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}$$

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L$$

Example 4.26. Let $f(x) = \sin(x)$ and $g(x) = x$ we have $\sin(0) = 0$ and $g(0) = 0$ further we have $f'(x) = \cos(x)$ and $g'(x) = 1 \neq 0$ hence

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \lim_{x \rightarrow 0} \frac{\cos(x)}{1} = 1$$

Remark 4.27. This example can also be solved via the definition of the derivative.

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \lim_{x \rightarrow 0} \frac{\sin(x) - \sin(0)}{x - 0} = \sin'(0) = \cos(0) = 1$$

4.3 Convex and Concave graphs

Definition 4.28. A function $f : I \rightarrow \mathbb{R}$ is called **convex** iff

$$\forall a, b \in I \forall t \in (0, 1) : f((1-t)a + bt) \leq (1-t)f(a) + tf(b)$$

Remark 4.29. This is equivalent to saying that for any $x \in (a, b)$ we have

$$\frac{f(x) - f(a)}{x - a} \geq \frac{f(b) - f(x)}{b - x}$$

Definition 4.30. A function $f : I \rightarrow \mathbb{R}$ is called **concave** iff

$$\forall a, b \in I \forall t \in (0, 1) : f((1-t)a + bt) \geq (1-t)f(a) + tf(b)$$

Theorem 4.31. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then

$$\forall x : f'(x) \geq 0 \iff f \text{ is increasing}$$

Remark 4.32. If $f'(x) = 0$ for all x then f is constant.

Theorem 4.33. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then

$$f \text{ is convex} \iff f' \text{ is increasing}$$

Proof. $\Leftarrow$: Assume f' is increasing and let $x \in (a, b)$ be arbitrary. Then we have by the mean value theorem that there exists $\psi \in (a, x)$ and $\xi \in (x, b)$ s.t.:

$$f'(\psi) = \frac{f(x) - f(a)}{x - a} \quad \text{and} \quad f'(\xi) = \frac{f(b) - f(x)}{b - x}$$

As f' is increasing we know that $f'(\psi) \leq f'(\xi)$ and therefore

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(x)}{b - x}$$

$\Rightarrow$: Assume f is convex and let h be small enough s.t. $x = a + h \leq b - h$ then by the convexity of f we have

$$\frac{f(a + h) - f(a)}{h} \leq \frac{f(b - h) - f(a + h)}{(b - h) - (a + h)} \leq \frac{f(b) - f(b - h)}{h}$$

and as f is differentiable they exist and are comparable therefore $f'(a) \leq f'(b)$ $\square$

Theorem 4.34. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable. Then

$$f \text{ is convex} \iff f'' \geq 0$$

4.4 Taylor Series

Lemma 4.35. We can locally approximate the graph of f at x_0 with the tangential line $t(x)$

$$f(x) \approx t(x) = f(x_0) + f'(x_0)(x - x_0)$$

Example 4.36. Let $f(x) = x^2$ and $x_0 = 2$ then

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) = 2^2 + 2(2)(x - 2) = 4 + 4(x - 2) = 4x - 4$$

Lemma 4.37. To even more accurately approximate the function f we can add further terms to the tangential line expansion

$$f(x) \approx t(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

Remark 4.38. The *Intuition* behind this is that we want a function that looks something like this:

$$P(x) = A + B(x - x_0) + C(x - x_0)^2$$

if $P(x_0) = f(x_0)$ we have $A = f(x_0)$. Further if $P'(x_0) = f'(x_0)$ we have $B = f'(x_0)$ and if $P''(x_0) = f''(x_0)$ we have $C = f''(x_0)/2$.

Example 4.39. Let $f(x) = x^2$ and $x_0 = 2$ then

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 = 2^2 + 2(2)(x - 2) + \frac{2}{2}(x - 2)^2 = 4 + 4(x - 2) + (x - 2)^2$$

Theorem 4.40. Taylor's Theorem: Let $f : (a, b) \rightarrow \mathbb{R}$ be $n+1$ times differentiable. Then

$$f(x) = P_n(x) + R(x)$$

with $P_n(x)$ the Taylor Polynomial of degree n and $R(x)$ the remainder term

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

$$R(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } \xi \in (x_0, x) \text{ or } (x, x_0)$$

Remark 4.41. There are other forms of the remainder term, such as the Lagrange form and the Cauchy form.

Example 4.42. Assume we know that the equation

$$\sqrt{3x - 4} + 2 \sin x - x = \frac{1}{2}$$

has a solution near $x_0 = 2$. We can use Taylor's Theorem to approximate the solution. Let $f(x) = \sqrt{3x - 4} + 2 \sin x - x$. Then

$$f(2) = \sqrt{2} + 2 \sin 2 - 2$$

$$f'(2) = \frac{3}{2\sqrt{2}} + 2 \cos 2 - 1$$

Therefore the Taylor Approximation up to degree 2 is

$$f(x) \approx f(2) + f'(2)(x - 2)$$

Setting this to $\frac{1}{2}$ and solving for x gives us the linear approximation of the solution.

Definition 4.43. The **Taylor series expansion** of an infinitely differentiable function $f : (a, b) \rightarrow \mathbb{R}$ at x_0 is given by

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

Example 4.44. The following Taylor series converge for all $x \in \mathbb{R}$:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Important 4.45. Not all infinitely differentiable functions have a Taylor series expansion that converges to the function itself. For example $\ln(1+x)$, $\frac{1}{1-x}$, $(1+x)^p$ only converge for $-1 < x < 1$

Theorem 4.46. Let $f(x)$ and $g(x)$ be two powerseries with x in their Radius of Convergence R .

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$g(x) = \sum_{n=0}^{\infty} b_n x^n$$

Then

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n$$

Theorem 4.47. Let $f(x)$ be a power series with x in its Radius of Convergence R .

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

then

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Theorem 4.48. Let $f(x)$ be a power series with x in its Radius of Convergence R .

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

then

$$\int f(x) dx = \sum_{n=0}^{\infty} \int a_n x^n dx$$

Example 4.49. Assume we know $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $-1 < x < 1$ and we want to calculate the Taylor-Expansion of $g(x) = (1-x)^{-2}$. We observe that $g(x) = (\frac{d}{dx})(1-x)^{-1}$. Therefore

$$g(x) = \frac{d}{dx} \sum_{n=0}^{\infty} x^n = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1) x^n$$

5 Integration

5.1 Indefinite Integrals and Techniques

Definition 5.1. Functions $F : I \rightarrow \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in (a, b)$ are called **antiderivatives** of f .

Definition 5.2. The family of all anti-derivatives of a function $f : I \rightarrow \mathbb{R}$ for $I \subseteq \mathbb{R}$ an open interval is called the **indefinite integral** and is denoted by

$$\int f(x)dx = \{F : I \rightarrow \mathbb{R} : F'(x) = f(x)\} = F(x) + C$$

Theorem 5.3. Let $f, g : I \rightarrow \mathbb{R}$ be integrable functions and let $c \in \mathbb{R}$. Then

$$\begin{aligned}\int (f(x) \pm g(x))dx &= \int f(x)dx \pm \int g(x)dx \\ \int cf(x)dx &= c \int f(x)dx\end{aligned}$$

Remark 5.4. Some important basic antiderivatives:

- $f(x) = 0 \implies F(x) = C$
- $f(x) = 1 \implies F(x) = x + C$
- $f(x) = x^p \implies F(x) = \frac{1}{p+1}x^{p+1} + C \quad (p \neq -1)$
- $f(x) = e^x \implies F(x) = e^x + C$
- $f(x) = a^x \implies F(x) = \frac{a^x}{\ln(a)} + C$
- $f(x) = \sin(x) \implies F(x) = -\cos(x) + C$
- $f(x) = \cos(x) \implies F(x) = \sin(x) + C$
- $f(x) = \frac{1}{x} \implies F(x) = \ln|x| + C$
- $f(x) = \frac{1}{1+x^2} \implies F(x) = \arctan(x) + C$

Theorem 5.5. Integration by Parts: For differentiable functions $f(x)$ and $g(x)$ we have

$$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx$$

Remark 5.6. The equation is derived from the product rule for differentiation.

Example 5.7. Let $f'(x) = e^x \implies f(x) = e^x$ and $g(x) = x \implies g'(x) = 1$. Then

$$\int x e^x dx = x e^x - \int e^x \cdot 1 dx = x e^x - e^x + C$$

Theorem 5.8. Substitution Rule: For functions $f(x)$ and $g(x)$ we have

$$\int f(g(x))g'(x)dx = \int f(u)u'dx = \int f(u)\frac{du}{dx}dx = \int f(u)du = F(g(x)) + C$$

where F is an antiderivative of f .

Remark 5.9. The equation is derived from the chain rule for differentiation.

Example 5.10. Evaluate $\int 2x e^{x^2} dx$. Let $u = x^2 \implies \frac{du}{dx} = 2x \implies du = 2x dx$. Substituting gives:

$$\int e^u du = e^u + C = e^{x^2} + C$$

5.2 The Definite Integral (Riemann Integral)

Remark 5.11. Upper and Lower Sums (Stair Functions): One can partition the integration interval into subintervals. For each subinterval we can approximate the area of the function with the area of a rectangle. The sum of the areas of these rectangles is an approximation of the area under the function. Taking the infimum of the function on each subinterval yields the lower sum L . Taking the supremum yields the upper sum U .

Definition 5.12. Let U and L denote the upper and lower sums of a bounded function f . Then f is called **Riemann-integrable** iff $\inf U = \sup L$. The value of the definite integral $\int_a^b f(x)dx$ is then this common value.

Theorem 5.13. Integrability Conditions:

- All continuous functions on $[a, b]$ are Riemann-integrable.
- All monotone functions on $[a, b]$ are Riemann-integrable.

Theorem 5.14. Let f and g be integrable and $c \in \mathbb{R}$ then:

- *Linearity:* $\int_a^b (f(x) \pm g(x))dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$
- *Monotonicity:* $f(x) \leq g(x) \implies \int_a^b f(x)dx \leq \int_a^b g(x)dx$
- *Triangle inequality:* $\left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx$
- *Inversion of bounds:* $\int_a^b f(x)dx = -\int_b^a f(x)dx$
- *Partition of interval:* $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$

Theorem 5.15. Mean Value Theorem for Integrals: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then there exists a $c \in [a, b]$ such that

$$\int_a^b f(x)dx = (b - a)f(c)$$

Theorem 5.16. Fundamental Theorem of Integral Calculus: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then for all $C \in \mathbb{R}$, $F(x)$ is an antiderivative of f where:

$$F(x) = \int_a^x f(y)dy + C$$

Proof. As f is continuous it is integrable on $[a, x]$ for any $x \in [a, b]$. Let $x_0 \in [a, b]$ and $\varepsilon > 0$ arbitrary. Then there exist a $\delta > 0$ such that for any $x \in [a, b]$ with $0 < |x - x_0| < \delta$ we have $|f(x) - f(x_0)| < \varepsilon$. Then for $x \in [a, b]$ with $0 < |x - x_0| < \delta$:

$$\begin{aligned} \left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| &= \left| \frac{1}{x - x_0} \left(\int_a^x f(y)dy - \int_a^{x_0} f(y)dy \right) - f(x_0) \right| \\ &= \left| \frac{1}{x - x_0} \int_{x_0}^x f(y)dy - \frac{1}{x - x_0} \int_{x_0}^x f(x_0)dy \right| = \left| \frac{1}{x - x_0} \int_{x_0}^x (f(y) - f(x_0))dy \right| \\ &\leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x |f(y) - f(x_0)|dy \right| \leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x \varepsilon dy \right| = \varepsilon \end{aligned}$$

□

Proof. Uniqueness of Antiderivatives: Let $G(x)$ be another antiderivative of $f : [a, b] \rightarrow \mathbb{R}$ where $F(x) = \int_a^x f(y)dy + C$. We define $H(x) = F(x) - G(x)$ and get

$$H'(x) = F'(x) - G'(x) = f(x) - f(x) = 0$$

Hence $H(x)$ must be a constant C' and $F(x) = G(x) + C'$. Thus, antiderivatives are unique up to a constant. □

5.3 Sequences of Functions and Integrals

Definition 5.17. Let $(f_n)_{n \in \mathbb{N}_0}$ be a sequence of functions.

- **Pointwise Convergence:** It converges pointwise to f if for every x , the sequence of real numbers $(f_n(x))$ converges to $f(x)$.
- **Uniform Convergence:** It converges uniformly to f if for every $\epsilon > 0$, there exists an N such that for all $n \geq N$ and **for all** x simultaneously, $|f_n(x) - f(x)| < \epsilon$.

Theorem 5.18. Swapping Limit and Integral: If a sequence of integrable functions (f_n) converges **uniformly** to f on $[a, b]$, then the limit function f is also integrable, and we can swap the limit and the integral:

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x)dx$$

5.4 Definite Integration Techniques

Lemma 5.19. If $F : [a, b] \rightarrow \mathbb{R}$ is continuously differentiable, then:

$$F(x) = F(a) + \int_a^x F'(t)dt \quad \text{and} \quad \int_a^b f(t)dt = F(b) - F(a) = \left[F(x) \right]_a^b$$

Lemma 5.20. Substitution for Definite Integrals: Let f be continuous and g continuously differentiable.

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

Version 2: If $f'(x) \neq 0$ and f^{-1} is the inverse of f :

$$\int_a^b g(f(x))dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y)dy$$

Example 5.21. Evaluate $\int_0^{\pi/2} \sin(x) \cos(x)dx$. Let $u = \sin(x)$. Then $\frac{du}{dx} = \cos(x) \implies du = \cos(x)dx$. For definite integrals, we must also substitute the bounds: Lower bound: $x = 0 \implies u = \sin(0) = 0$. Upper bound: $x = \pi/2 \implies u = \sin(\pi/2) = 1$. Substituting gives:

$$\int_0^1 u du = \left[\frac{1}{2}u^2 \right]_0^1 = \frac{1}{2}(1)^2 - \frac{1}{2}(0)^2 = \frac{1}{2}$$

Lemma 5.22. *Integration by Parts for Definite Integrals:* Let f, g be continuously differentiable.

$$\int_a^b f'(x)g(x)dx = \left[f(x)g(x) \right]_a^b - \int_a^b f(x)g'(x)dx$$

Example 5.23. Evaluate $\int_1^e \ln(x)dx$. We can write this as $\int_1^e 1 \cdot \ln(x)dx$. Let $f'(x) = 1 \implies f(x) = x$. Let $g(x) = \ln(x) \implies g'(x) = \frac{1}{x}$. Using integration by parts for definite integrals:

$$\begin{aligned} \int_1^e 1 \cdot \ln(x)dx &= \left[x \ln(x) \right]_1^e - \int_1^e x \cdot \frac{1}{x}dx \\ &= (e \ln(e) - 1 \ln(1)) - \int_1^e 1dx = e - \left[x \right]_1^e = e - (e - 1) = 1 \end{aligned}$$

6 Differential Equations

6.1 Modelling and Definitions

Example 6.1. Assume at time $t = 0$ one takes in a certain amount of alcohol. The alcohol first flows to the stomach and is then absorbed into the blood. We want to find a description of the current amount of alcohol in the stomach $m_S(t)$ and in the blood $m_B(t)$.

We assume the following:

- the amount of alcohol in the stomach decreases by $\alpha\Delta t \cdot m_S(t)$

$$\begin{aligned}m_S(t + \Delta t) &= m_S(t) - \alpha\Delta t \cdot m_S(t) \\ \implies \frac{m_S(t + \Delta t) - m_S(t)}{\Delta t} &= -\alpha \cdot m_S(t) \\ \implies m'_S(t) &= -\alpha \cdot m_S(t)\end{aligned}$$

- the amount of alcohol in the blood is decreased by a constant amount of $\beta\Delta t$

$$m'_B(t) = \alpha \cdot m_S(t) - \beta$$

Example 6.2. A differential equation describing bounded growth, i.e., a quantity that cannot grow arbitrarily large, is modelled as:

$$u'(t) = ku(t) \left(1 - \frac{u(t)}{L}\right)$$

where L and k are positive constants. The value L represents the maximum capacity. This equation can be simplified into a separable differential equation using the substitution $v(t) = \frac{1}{u(t)}$.

Example 6.3. A differential equation describing a bimolecular reaction $A + B \rightarrow C$, where $u(t)$ is the amount of substance C at time t , is given by:

$$u'(t) = k(a - u(t))(b - u(t))$$

where a, b , and k are positive real constants representing the initial concentrations and the reaction rate, respectively.

Definition 6.4. A differential equation is an equation which contains derivatives of functions.

Definition 6.5. A differential equation is called

- of ***n*-th order**, if the *n*-th derivative is the highest occurring derivative.
- **linear**, if the unknown function $u(t)$ and its derivatives appear in the equation in the form of a linear combination.
- **homogeneous**, if a linear ODE only contains multiples of the unknown function $u(t)$ and its derivatives (i.e. right hand side is 0). It is **inhomogeneous** if

there is a term (the disturbance function) that only depends on the independent variable.

Definition 6.6. An ordinary differential equation (ODE) is a differential equation which contains only derivatives of one independent variable.

Theorem 6.7 (Picard-Lindelöf - Existence and Uniqueness). If $f(x, y)$ is continuous and Lipschitz continuous with respect to y , then the initial value problem $y' = f(x, y)$ with $y(x_0) = y_0$ has a unique solution in some interval around x_0 .

6.2 First-Order ODEs

Theorem 6.8. For an ODE of the form $y' = f(x)g(y)$, we can separate the variables and integrate:

$$\frac{dy}{dx} = f(x)g(y) \implies \int \frac{1}{g(y)} dy = \int f(x) dx + C$$

Example 6.9. Solve $y' = 2xy$. We separate: $\frac{1}{y} dy = 2x dx$. Integrating both sides gives $\ln |y| = x^2 + C$. Solving for y yields $y = \pm e^C e^{x^2} = K e^{x^2}$.

Theorem 6.10. If a differential equation is of the form $y' = f(ax + by + c)$, we can substitute $u = ax + by + c$. This transforms the equation into an autonomous differential equation with separable variables.

Example 6.11. Solve $y' = (x + y)^2$. Substitute $u = x + y$, then $u' = 1 + y'$. The ODE becomes $u' - 1 = u^2 \implies u' = u^2 + 1$. This can now be solved using separation of variables: $\int \frac{1}{u^2+1} du = \int 1 dx \implies \arctan(u) = x + C$. Back-substituting gives $x + y = \tan(x + C) \implies y = \tan(x + C) - x$.

Theorem 6.12. If a differential equation can be written in the form:

$$y' = g\left(\frac{y}{x}\right)$$

it is called homogeneous in the variables. We can solve this by substituting $u = \frac{y}{x}$, which transforms it into an equation with separable variables.

Example 6.13. Solve the differential equation $y' = \frac{x^2+y^2}{xy}$.

First, we rewrite the right-hand side to show it is homogeneous in the variables (by dividing each term by xy):

$$y' = \frac{x^2}{xy} + \frac{y^2}{xy} = \frac{x}{y} + \frac{y}{x} = \frac{1}{\frac{y}{x}} + \frac{y}{x}$$

We use the substitution $u = \frac{y}{x}$, which implies $y = u \cdot x$. By the product rule, the derivative is $y' = u'x + u$.

Substituting y' and $\frac{y}{x}$ into our equation yields:

$$u'x + u = \frac{1}{u} + u$$

Subtracting u from both sides gives a separable differential equation:

$$u'x = \frac{1}{u} \implies u \, du = \frac{1}{x} \, dx$$

Integrating both sides:

$$\int u \, du = \int \frac{1}{x} \, dx \implies \frac{1}{2}u^2 = \ln|x| + C$$

Solving for u :

$$u^2 = 2 \ln|x| + 2C \implies u = \pm \sqrt{2 \ln|x| + K}$$

Finally, back-substituting $u = \frac{y}{x}$:

$$\frac{y}{x} = \pm \sqrt{2 \ln|x| + K} \implies y = \pm x \sqrt{2 \ln|x| + K}$$

6.3 Linear ODEs of n-th Order

Theorem 6.14. If y_1 and y_2 are solutions to a linear homogeneous differential equation so is

$$y(x) = C_1 y_1(x) + C_2 y_2(x), \quad C_1, C_2 \in \mathbb{R}$$

Theorem 6.15. Linear homogenous differential equations of n -th order have n linearly independent solutions $y_1, \dots, y_n$ such that

$$y(x) = C_1 y_1(x) + \dots + C_n y_n(x), \quad C_i \in \mathbb{R}, \text{ for } i = 1, \dots, n$$

is called the general solution.

Lemma 6.16. Let $y(x) = e^{\lambda x}$, $\lambda \in \mathbb{C}$. By deriving n times and substituting into the homogeneous ODE with constant coefficients, we obtain the characteristic equation:

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0 = 0$$

The corresponding polynomial is the characteristic polynomial $p(\lambda)$.

Theorem 6.17. We can find the general solution of a linear, homogeneous ODE of n -th order with constant coefficients by finding the roots of the characteristic polynomial:

- **Distinct real roots:** For each distinct root $\lambda_i \in \mathbb{R}$, we get a solution $y_i(x) = e^{\lambda_i x}$.
- **Multiple real roots:** If a root $\lambda_i \in \mathbb{R}$ has multiplicity m , the m linearly independent solutions are $e^{\lambda_i x}, x e^{\lambda_i x}, x^2 e^{\lambda_i x}, \dots, x^{m-1} e^{\lambda_i x}$.
- **Complex roots:** Complex roots always come in conjugate pairs $\lambda = \alpha \pm i\beta \in \mathbb{C}$. If they have multiplicity 1, they yield two real solutions: $y_1(x) = e^{\alpha x} \cos(\beta x)$ and $y_2(x) = e^{\alpha x} \sin(\beta x)$.

Example 6.18. Solve $y'' - 5y' + 6y = 0$. Using the Euler Ansatz $y(x) = e^{\lambda x}$, we obtain the characteristic equation:

$$\lambda^2 - 5\lambda + 6 = 0$$

The roots are $\lambda_1 = 2$ and $\lambda_2 = 3$. Both are distinct real roots. The general solution is therefore $y(x) = C_1 e^{2x} + C_2 e^{3x}$.

6.4 Inhomogeneous Linear ODEs

Lemma 6.19. To solve general linear ODEs, we first find the general solution of the associated linear homogeneous ODE $y_h(x)$ of n -th order. Then we use an appropriate form $y_p(x)$ to find the particular solution of the inhomogeneous ODE. The general solution is then $y(x) = y_h(x) + y_p(x)$.

6.4.1 Method of Undetermined Coefficients (Ansatz)

This method works for ODEs with **constant coefficients** and specific types of right-hand sides (disturbance functions) $s(x)$.

Theorem 6.20. If $s(x) = a_0 + a_1 x + \dots + a_m x^m$, then let $y_p(x) = C_0 + C_1 x + \dots + C_m x^m$. Find the constants C_i by plugging $y_p(x)$ into the ODE.

Theorem 6.21. If $s(x) = A e^{kx}$, then let $y_p(x) = C e^{kx}$. Find C by plugging $y_p(x)$ into the ODE.

Theorem 6.22 (Trigonometric / Oscillation). If $s(x) = A \sin(\omega x) + B \cos(\omega x)$, then let $y_p(x) = C_1 \sin(\omega x) + C_2 \cos(\omega x)$. Find the constants C_i by plugging $y_p(x)$ into the ODE. Note: You must always include both sine and cosine in the Ansatz!

Important 6.23. Resonance (Special Cases): If the structure of the right-hand side $s(x)$ matches a root of the characteristic polynomial (e.g. k is an m -fold root, or $i\omega$ is an m -fold root), then we must multiply our standard Ansatz by x^m . For example, if $s(x) = Ae^{kx}$ and k is a single root ($m = 1$) of $p(\lambda)$, the Ansatz becomes $y_p(x) = x \cdot Ce^{kx}$.

Example 6.24 (Undetermined Coefficients). Find a particular solution for $y'' - 3y' + 2y = e^{3x}$. The right-hand side is an exponential $s(x) = e^{3x}$, with $k = 3$. The roots of the characteristic polynomial are $\lambda_1 = 1, \lambda_2 = 2$. Since $k = 3$ is not a root, we have no resonance. We use the Ansatz $y_p(x) = Ce^{3x}$. Plugging it into the ODE gives:

$$y_p''(x) - 3y_p'(x) + 2y_p(x) = 9Ce^{3x} - 3(3Ce^{3x}) + 2Ce^{3x} = 2Ce^{3x}$$

We equate this to the right-hand side e^{3x} :

$$2Ce^{3x} = e^{3x} \implies 2C = 1 \implies C = \frac{1}{2}$$

The particular solution is $y_p(x) = \frac{1}{2}e^{3x}$.

Example 6.25 (Resonance). Find a particular solution for $y'' - 3y' + 2y = e^{2x}$. The right-hand side is $s(x) = e^{2x}$, with $k = 2$. The roots of the characteristic polynomial are $\lambda_1 = 1, \lambda_2 = 2$. Since $k = 2$ matches a root, we have resonance ($m = 1$). Our Ansatz must be multiplied by x : $y_p(x) = x \cdot Ce^{2x}$. We plug $y_p(x)$, $y_p'(x) = C(1 + 2x)e^{2x}$, and $y_p''(x) = C(4 + 4x)e^{2x}$ into the ODE to solve for C :

$$C(4 + 4x)e^{2x} - 3C(1 + 2x)e^{2x} + 2Cxe^{2x} = e^{2x}$$

Dividing by e^{2x} gives: $4C + 4Cx - 3C - 6Cx + 2Cx = 1 \implies C = 1$. The particular solution is $y_p(x) = xe^{2x}$.

6.4.2 Variation of Constants (First-Order ODEs)

Theorem 6.26 (Variation of Constants). For an inhomogeneous, linear differential equation of first order, we can find the particular solution by replacing the constant C from the homogeneous solution with a function $C(x)$.

If the homogeneous solution is $y_h(x) = C \cdot y_1(x)$, we set the particular solution to:

$$y_p(x) = C(x) \cdot y_1(x)$$

By plugging this $y_p(x)$ back into the original inhomogeneous ODE, we get a condition for $C'(x)$, which we can integrate to find the function $C(x)$.

6.5 Initial Value Problems and Boundary-Value Problems

The general solution $y(x)$ of an n -th order ODE contains n unknown constants. We need n additional conditions to specify a unique solution curve.

6.5.1 Initial Value Problem (IVP)

An Initial Value Problem specifies the conditions at the **same point** (e.g., t_0).

Example 6.27. Solve the IVP: $y'' + y = 0$ with $y(0) = 1$ and $y'(0) = 0$. The general solution is $y(x) = C_1 \cos(x) + C_2 \sin(x)$. Using the conditions: $y(0) = C_1 \cdot 1 + C_2 \cdot 0 = 1 \implies C_1 = 1$. $y'(x) = -C_1 \sin(x) + C_2 \cos(x) \implies y'(0) = C_2 \cdot 1 = 0 \implies C_2 = 0$. The specific solution is $y(x) = \cos(x)$.

6.5.2 Boundary-Value Problem (BVP)

A Boundary-Value Problem specifies the conditions at **different points** (often boundaries of an interval, $t_0 \neq t_1$).

Example 6.28. Solve the BVP: $y'' + y = 0$ with $y(0) = 0$ and $y(\pi/2) = 1$. The general solution is $y(x) = C_1 \cos(x) + C_2 \sin(x)$. Using the first boundary condition: $y(0) = C_1 \cdot 1 + 0 = 0 \implies C_1 = 0$. Using the second boundary condition: $y(\pi/2) = 0 + C_2 \cdot 1 = 1 \implies C_2 = 1$. The specific solution is $y(x) = \sin(x)$.